

URBANFIX: AN AI-POWERED END-TO-END GOVERNANCE SYSTEM FOR CIVIC INFRASTRUCTURE COMPLAINTS

A PROJECT REPORT

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BONAFIDE CERTIFICATE

Certified that this project report titled "**UrbanFix: An AI-Powered End-to-End Governance System for Civic Infrastructure Complaints**" is the bonafide work of **Parth Pancholi (Register No: 25BCE10443)**, **Aditya Dev (Register No: 25BCE10863)**, **Prince Mahar (Register No: 25BCE10429)**, **Arun Kumar (Register No: 25BCE11009)**, **Aditi Sahu (Register No: 25BCE10844)** who carried out the project work under my supervision. Certified further that to the best of my knowledge the work reported at this time does not form part of any other project or research work based on which a degree or award was conferred on an earlier occasion on this or any other candidate.

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The Project Exhibition I Examination is held on _____

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Finally, we thank our families and friends for their patience and support.

Parth Pancholi, Aditya Dev, Prince Mahar, Arun Kumar, Aditi Sahu

LIST OF ABBREVIATIONS

Term	Expansion	Term	Expansion
AI	Artificial Intelligence	MPS	Metal Performance Shaders (Apple GPU backend)
API	Application Programming Interface	NLP	Natural Language Processing
CPGRAMS	Centralised Public Grievance Redress and Monitoring System	NSFW	Not Safe For Work
CPU	Central Processing Unit	OCR	Optical Character Recognition
CRDDC	Crowdsensing-based Road Damage Detection Challenge	OTP	One-Time Password
CSV	Comma-Separated Values	PDF	Portable Document Format
DARPG	Department of Administrative Reforms and Public Grievances	RDD	Road Damage Dataset
ER	Entity-Relationship	REST	Representational State Transfer
FY	Financial Year	SMTP	Simple Mail Transfer Protocol
GPU	Graphics Processing Unit	TACO	Trash Annotations in Context
HMAC	Hash-based Message Authentication Code	TF-IDF	Term Frequency, Inverse Document Frequency
HTTPS	Hypertext Transfer Protocol Secure	TTL	Time To Live
JSON	JavaScript Object Notation	UI	User Interface
JWT	JSON Web Token	UT	Union Territory
mAP	mean Average Precision	YOLO	You Only Look Once (object detector family)
MoRTH	Ministry of Road Transport and Highways		

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ABSTRACT

Purpose. Civic complaints in India are slow to resolve and hard to follow. In FY 2023-24, State and UT governments took 119 days on average to dispose of a grievance on CPGRAMS, and in 2024 potholes were linked to 5,432 road accidents. UrbanFix is a mobile governance system that follows a complaint from the citizen's photo to a checked and publicly recorded repair.

Methodology. Citizens report without an account and confirm each report with an email OTP. A YOLOv8 model trained for the chosen category checks the photo, and a weighted questionnaire sets an urgency level that anyone can trace back to the answers. The complaint then moves through seven stages enforced by the server, from supervisor triage and assignment within the ward to field repair, photo proof and closure with a PDF receipt. Every status change is public, and approved researchers can query anonymised data for a limited time. The app offers only the four categories with a public annotated dataset: potholes, litter, open manholes and graffiti.

Findings. On held-out data the pothole model reaches a mask mAP50 of 0.724, the litter model a mask mAP50 of 0.502, the open manhole model a box mAP50 of 0.908 and the graffiti model a box mAP50 of 0.729. All 16 API validation tests pass.

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CHAPTER 1

PROJECT DESCRIPTION AND OUTLINE

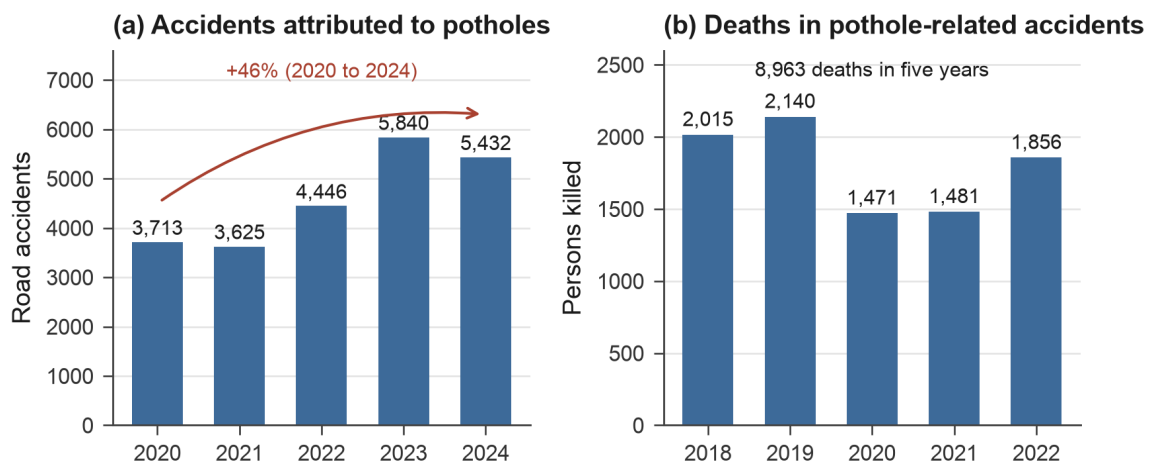
1.1 Introduction

Potholes, uncollected garbage, open manholes and graffiti are easy to fix one at a time. Getting them fixed is harder: someone has to hear about the fault, confirm it, judge its urgency, send a worker, check the repair and tell the person who reported it. Today these steps are split across phone lines, office visits, social media and state portals, and the person who complained rarely hears back.

UrbanFix puts these steps in one AI-powered system: an Expo (React Native) app, an Express and MongoDB API, and a Python service running YOLOv8 models. Each complaint is tracked from the first photo to the final receipt in a public registry.

1.2 Motivation for the Work

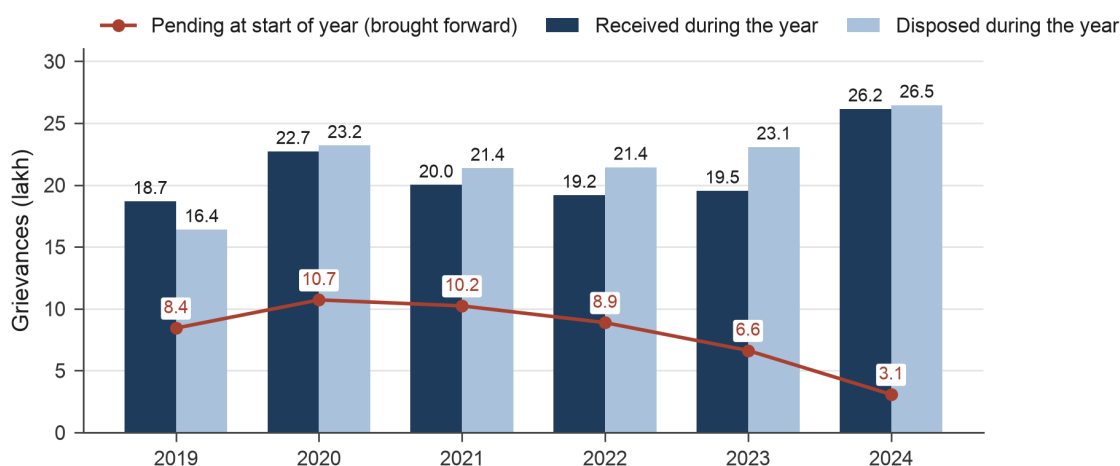
Potholes caused 5,432 road accidents in 2024, up from 3,713 in 2020, according to data the Ministry of Road Transport and Highways gave the Lok Sabha in February 2026. Its annual reports count 8,963 deaths in pothole-related accidents from 2018 to 2022 (Fig. 1.1).



Sources: (a) MoRTH data laid in the Lok Sabha, 12 Feb 2026. (b) MoRTH, Road Accidents in India 2018 to 2022.

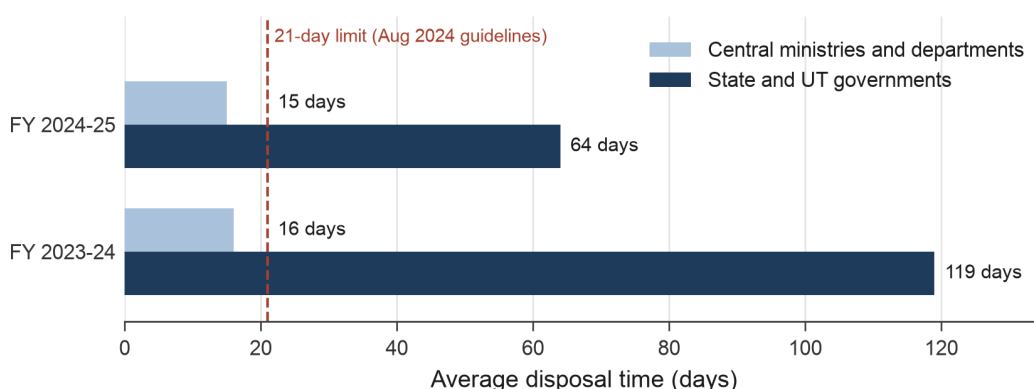
Fig. 1.1 Pothole-related road accidents (2020 to 2024) and deaths (2018 to 2022) in India

CPGRAMS, the national grievance portal, received 18.7 to 26.2 lakh new grievances a year from 2019 to 2024, and its backlog fell from 10.7 lakh to 3.1 lakh (Fig. 1.2). Speed depends on who handles the case: in FY 2023-24 central ministries took 16 days on average and State and UT governments 119 days (Fig. 1.3). Municipal complaints go to the state side.



Source: DARPG, CPGRAMS figures in Parliament replies (PIB, 16 Mar 2022 and 2 Apr 2025). 1 lakh = 100,000.

Fig. 1.2 Grievances received and disposed on CPGRAMS, with the backlog at the start of each year (2019 to 2024)



Source: DARPG reply in the Rajya Sabha on the CPGRAMS portal (PIB, 5 Feb 2026). Municipal and civic complaints fall under State/UT governments.

Fig. 1.3 Average grievance disposal time on CPGRAMS: central ministries against State and UT governments

1.3 About the Project and Techniques Used

UrbanFix works as a loop of six steps: Report, Validate, Prioritise, Act, Prove and Learn (Fig. 4.1).

- Citizens report through a six-step wizard and confirm with a six-digit email code instead of an account.
- A YOLOv8 model per category boxes or masks the defect and returns a confidence; unsure results go to a person.
- Weighted yes/no answers set the urgency, with the model's damage estimate as a second signal.
- The server enforces seven stages, assignment within the ward, and photo proof before closure.

- Every complaint and its history is public without the citizen's identity, and researchers get logged access to anonymised data.

1.4 Problem Statement

Civic complaints arrive through scattered channels, are checked by hand, prioritised by no visible rule and resolved out of public view. The task is to build one system that checks each report against photo evidence, ranks it by an urgency rule anyone can follow, moves it through an accountable workflow and publishes the result, without asking citizens to create an account.

1.5 Objectives of the Work

- Let citizens report from a phone with an email OTP instead of an account.
- Check every photo with a model trained on a public dataset for its category.
- Compute an urgency level that can be traced back to the answers.
- Enforce a ward workflow with role checks, remarks, photo proof and an audit trail.
- Publish every complaint and give researchers anonymised data.

1.6 Organization of the Report

Chapter 2 reviews related systems and datasets, Chapter 3 the requirements and Chapter 4 the design. Chapter 5 covers implementation, screens, training and tests, Chapter 6 outcomes, and Chapter 7 limitations and next steps.

1.7 Summary

Grievances are many, local redress is slow and potholes cause thousands of road deaths. UrbanFix answers with account-free reporting, AI photo checks, a traceable urgency rule, an enforced ward workflow and a public record.

CHAPTER 2

RELATED WORK INVESTIGATION

2.1 Introduction

We reviewed seven IEEE conference papers from 2025 and 2026 on AI-assisted grievance systems ([1] to [7]) and the public image datasets for the defects UrbanFix handles.

2.2 Core Area of the Project

A grievance system takes in a complaint, routes it, decides its urgency and reports the outcome. Computer vision adds one more job: confirming that the photo shows the reported defect.

2.3 Existing Approaches and Methods

2.3.1 Web portals with location and image evidence

Infra-Alert [1] tags complaints with the Geolocation API, stores before and after photos in Cloudinary and generates PDF reports. AI verification and prioritisation are left for future work, and citizens see only their own complaints, as on CPGRAMS.

2.3.2 NLP-based routing and urgency

CitiWatch [2] routes complaints with TF-IDF and a supervised classifier (96.8%). VidyutSaathi [3] uses Sentence-BERT, a multi-task network and DBSCAN duplicate grouping, and reports 81% faster response. CivicX [6] reaches 91.4% against 94.1% for officers, and Smart City Concern [7] uses zero-shot BART (92.8%) with sentiment as a stand-in for urgency. Text routing now sits between 90% and 97% (Fig. 2.3).

2.3.3 Image handling and defect datasets

ReGenX [4] sorts waste photos into four types (85.2%) and CivicMate [5] screens photos for NSFW content (F1 0.97); neither checks that the photo shows the reported defect. Datasets that would allow such a check exist for potholes (RDD2022 [9]), litter (TACO [10]), open manholes (Road Hazards [11]) and graffiti (STORM [12]).

2.4 Pros and Cons of the Stated Approaches

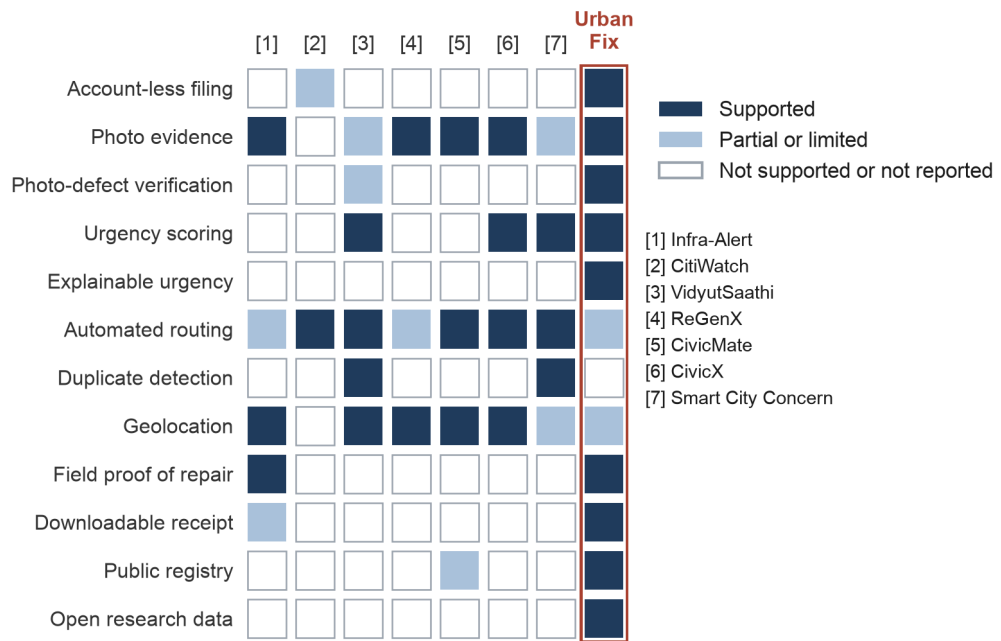
Table 2.1 Summary of the reviewed systems

System	Main method	Strength	Limitation
[1] Infra-Alert	Geolocation API, Cloudinary, PDF reports	Before and after proof, location accuracy	No AI verification or urgency; own-complaints view only
[2] CitiWatch	TF-IDF + supervised classifier, phone OTP	96.8% routing accuracy	No images; no public registry
[3] VidyutSaathi	Sentence-BERT, multi-task NN, DBSCAN	Urgency, duplicates and workload in one pipeline	Opaque urgency; internal tool only
[4] ReGenX	Geofencing, on-device TFLite	Low cost, no IoT hardware	Waste domain only; no public view
[5] CivicMate	NudeNet NSFW filter, CNN, chatbot	Accurate content moderation	Checks safety, not defect authenticity
[6] CivicX	RF/SVM/transformer, geo-heatmaps	Benchmarked against officers	Official-only transparency
[7] Smart City Concern	Zero-shot BART, DistilBERT, OCR	Web and social intake, duplicates	Sentiment stands in for urgency

Fig. 2.1 rates each system on twelve capabilities, and Fig. 2.2 counts how many of the seven lack each one.

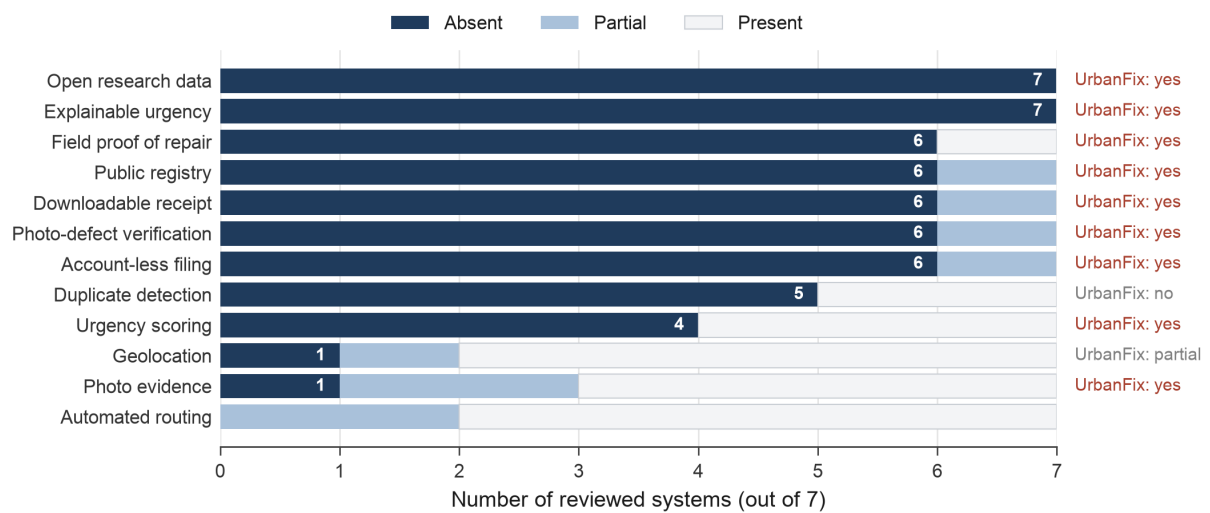
2.5 Issues and Observations from the Investigation

- None checks that a photo shows the reported defect (Fig. 2.2).
- None publishes every complaint with its history; CivicMate's community view is partial.
- Urgency comes from opaque models ([3], [6]) or sentiment ([7]).
- Text routing is close to solved.
- Usable datasets exist for only some complaint types (Section 3.3.1).



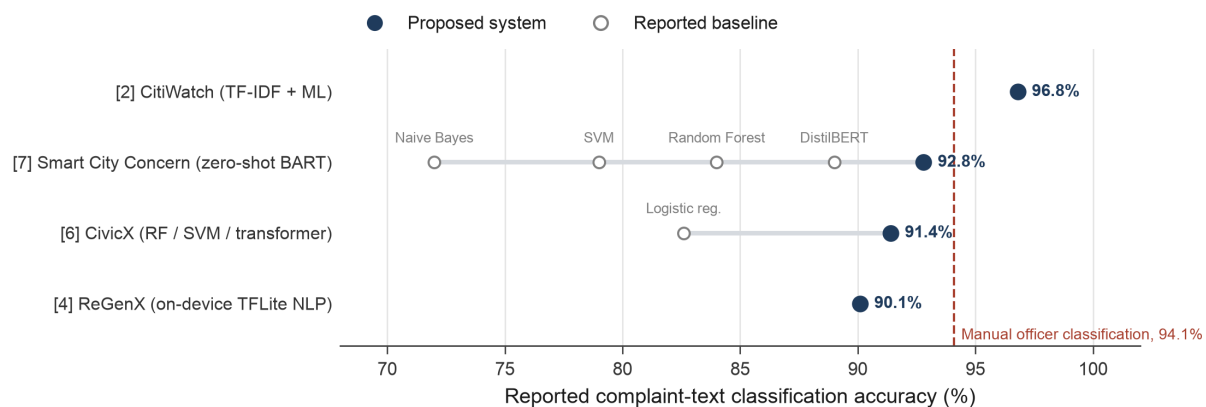
Source: authors' literature survey of [1] to [7]; ratings follow each paper's reported implementation.

Fig. 2.1 Feature coverage of the reviewed systems [1] to [7] compared with UrbanFix



Derived from the feature coverage matrix (Fig. 2.1).

Fig. 2.2 Number of reviewed systems lacking each capability



Source: results reported in [2], [4], [6] and [7]; manual baseline from [6].

Fig. 2.3 Reported complaint-text classification accuracy in the literature, with baselines and the manual officer reference

2.6 Summary

The reviewed systems route text well but do not check photos, explain urgency or publish their records. UrbanFix covers these points for the categories that public data supports.

CHAPTER 3

REQUIREMENT ARTIFACTS

3.1 Introduction

This chapter lists what UrbanFix needs to run and what each role must be able to do.

3.2 Hardware and Software Requirements

Table 3.1 Hardware and software requirements

Component	Requirement used in this project
Development machine	Apple M5 laptop, 16 GB unified memory, macOS; Apple MPS backend for model training
Phone (client)	Android or iOS phone with Expo Go supporting SDK 57; same Wi-Fi network as the server during development
Mobile app	Expo SDK 57, React Native 0.86, React 19, React Navigation 7, Reanimated 4, FlashList, react-native-svg, Axios
Backend	Node.js 20 or newer, Express 4, Mongoose 8, JSON Web Tokens, bcrypt, Zod, Multer, Nodemailer, PDFKit
Database	MongoDB 8.3 (any version 7 or newer works), local instance
AI service	Python 3, FastAPI, PyTorch 2.14, Ultralytics 8.4 (YOLOv8), Jupyter for training notebooks
Email	Any SMTP server; Ethereal test inbox used automatically in development

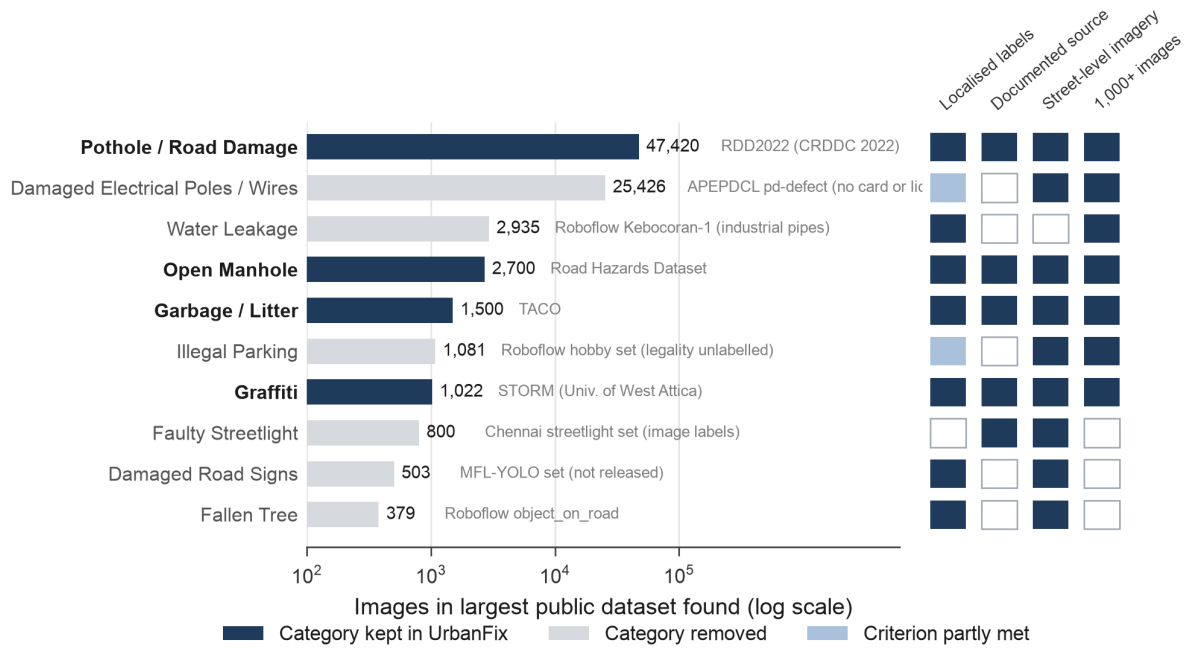
3.3 Specific Project Requirements

3.3.1 Data requirement

A category is offered only if a public, documented dataset of street-level images with boxes or masks exists for it. Of the ten categories in the original design, six failed (Fig. 3.1): water leakage, illegal parking and fallen trees have only small hobby sets, the streetlight set is labelled per image, the road sign set was never released and the electrical pole set has no licence.

Table 3.2 Training datasets for the four UrbanFix categories

Category	Dataset	Images	Labels	Split used
Pothole / Road Damage	Roboflow Pothole Segmentation YOLOv8 [8]	780	Polygon masks	720 train / 60 val
Garbage / Litter	TACO [10]	1500	COCO masks, merged to one class	1214 train / 214 val
Open Manhole	Road Hazards Dataset [11]	2,716	YOLO boxes (manhole class)	1123 train / 480 val
Graffiti	STORM graffiti/tagging dataset [12]	1,022	Bounding boxes	813 train / 209 test



Sources: RDD2022, TACO, Road Hazards Dataset, STORM, Hugging Face, Roboflow Universe and arXiv:2407.01117. Sizes are image counts as published.

Fig. 3.1 Largest public dataset found for each candidate complaint category and the selection criteria it meets

3.3.2 Functional requirements

Table 3.3 Functional requirements by role

Role	Functions
Citizen (no account)	File complaint with category, questions, ward, photos; verify by email OTP; receive tracking ID; track any complaint; download PDF receipt; browse registry
Supervisor	Triage (accept or reject with remarks); assign field worker in ward; review proof; close or send for rework; approve field leave; own attendance and leave
Field worker	View assigned tasks; start work; upload up to three proof photos with a completion note; check in and out; request leave
Admin	Analytics dashboard; status override; create, edit, deactivate staff through invite links; approve supervisor leave; approve research access; view research audit log
Researcher	Apply publicly; after approval, view insights, run grouped queries and export CSV or JSON within scope and limits
AI service	Run the category model on complaint photos; return detections, confidence, annotated images and damage share

3.3.3 Performance and security requirements

- The tracking ID returns at once; AI analysis runs afterwards and cannot block a submission.
- OTPs expire in five minutes, tokens in seven days; passwords use bcrypt (cost 12).
- Staff set passwords through 72-hour invite links stored only as SHA-256 hashes.
- Every role route checks the JWT and role; workflow changes follow one transition table.
- Public responses never include citizen IDs.
- Research data passes through one anonymising projection; exports are capped at 5 a day and 5,000 rows.

3.3.4 Look and feel requirements

The app uses a light theme with one pink accent (#FF5FA2), Space Grotesk and Plus Jakarta Sans, rounded cards and a glass tab bar. Status colours are fixed: amber Pending, violet In Progress, green Resolved, red Rejected.

3.4 Summary

UrbanFix runs on ordinary hardware with open-source software, and its categories follow from the data requirement.

CHAPTER 4

DESIGN METHODOLOGY AND ITS NOVELTY

4.1 Methodology and Goal

Each complaint passes through six steps (Fig. 4.1), and each leaves a record the next one uses, so nothing closes without evidence.



Fig. 4.1 The UrbanFix governance loop: Report, Validate, Prioritise, Act, Prove and Learn

We built the citizen flow first, then the staff workflow, the research layer and the AI service, testing on Android phones through Expo Go.

4.2 Functional Modules Design and Analysis

Fig. 4.2 maps each actor to their functions.

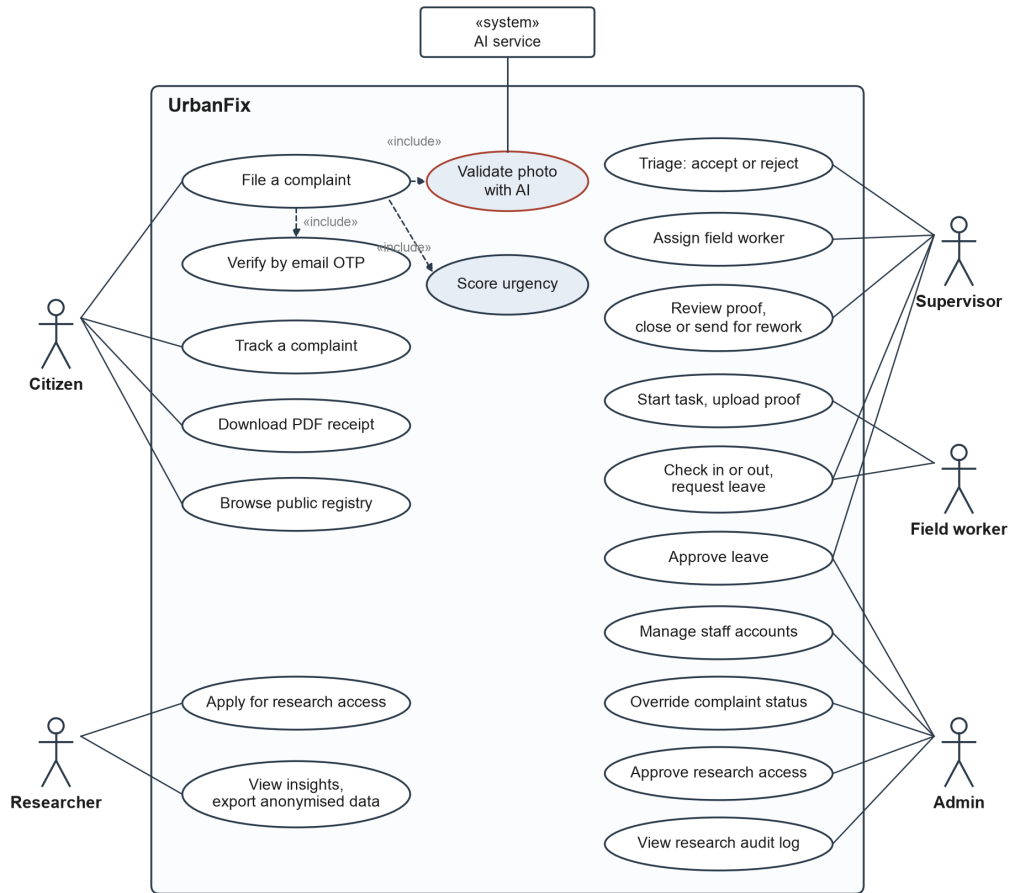


Fig. 4.2 Use-case diagram of UrbanFix by role

Staff see a detailed stage and citizens a simpler status (Fig. 4.3). Permitted moves are defined in `backend/utils/complaintStage.js` (Table 4.1).

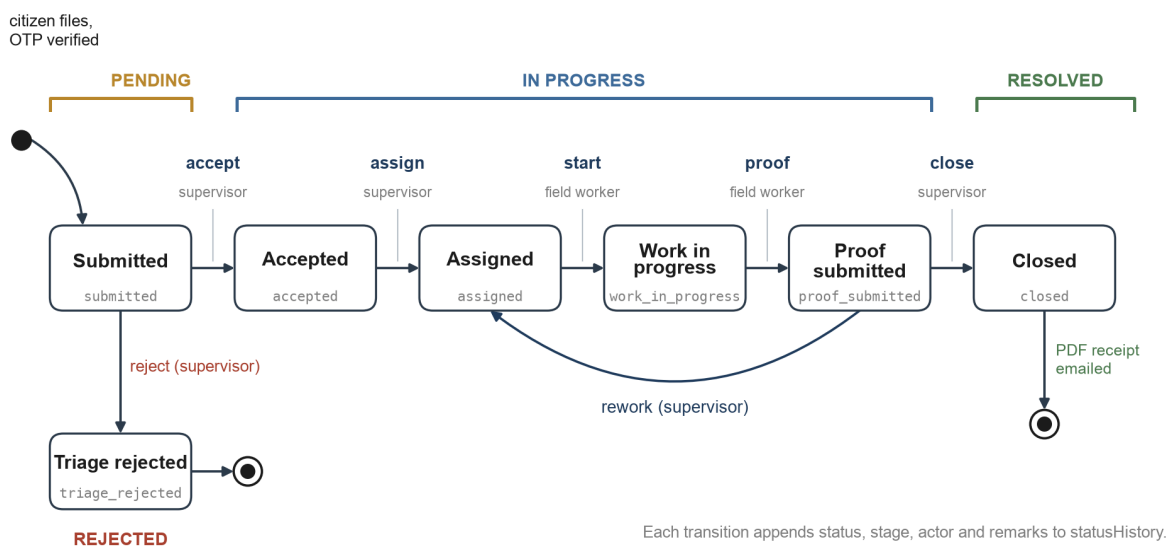
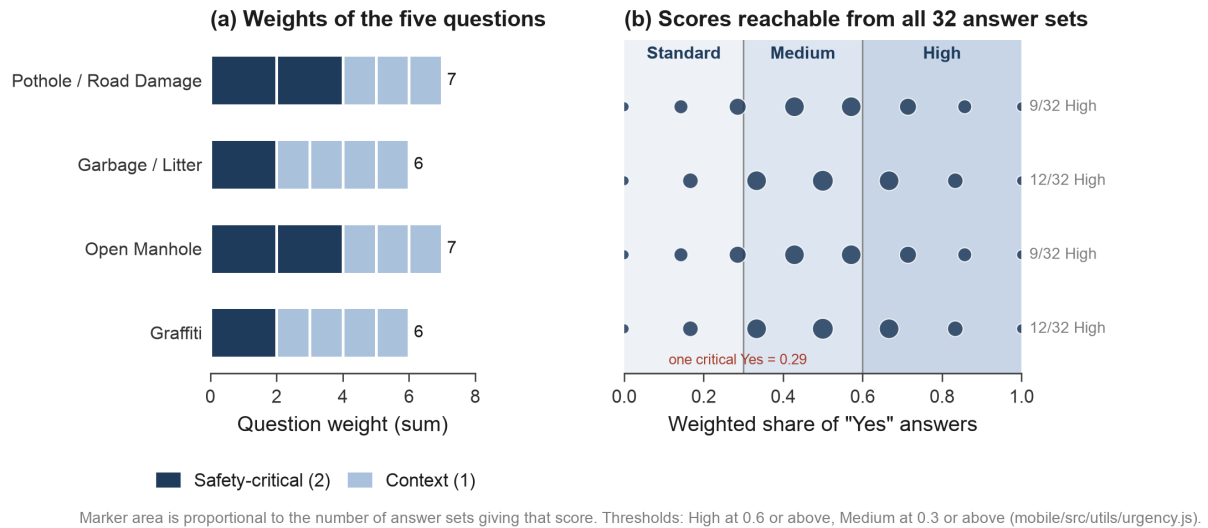


Fig. 4.3 Complaint lifecycle: internal stages, public statuses and allowed transitions

Table 4.1 Workflow transitions enforced by the server

Action	From	To	Who	Extra rule
accept	submitted	accepted	supervisor, admin	remarks of 10+ characters
reject	submitted	triage_rejected	supervisor, admin	remarks of 10+ characters
assign	accepted, assigned	assigned	supervisor, admin	same ward; worker active and not on leave
start	assigned	work_in_progress	assigned field worker	only the assignee
proof	work_in_progress	proof_submitted	assigned field worker	note and up to 3 photos
close	proof_submitted	closed	supervisor, admin	remarks; PDF receipt emailed
rework	proof_submitted	assigned	supervisor, admin	remarks of 10+ characters

Each category has five questions, weighted 2 for safety and 1 otherwise. The weighted share of "Yes" answers gives High at 0.6, Medium at 0.3, and Standard below (Fig. 4.4). For a pothole, one safety "Yes" scores 0.29 and two score 0.57.

**Fig. 4.4** Urgency scoring: question weights per category and all reachable scores

4.3 Software Architectural Designs

UrbanFix has three tiers and a separate AI service (Fig. 4.5). The Express API holds the rules, MongoDB stores seven collections, and only the backend calls the FastAPI model service.

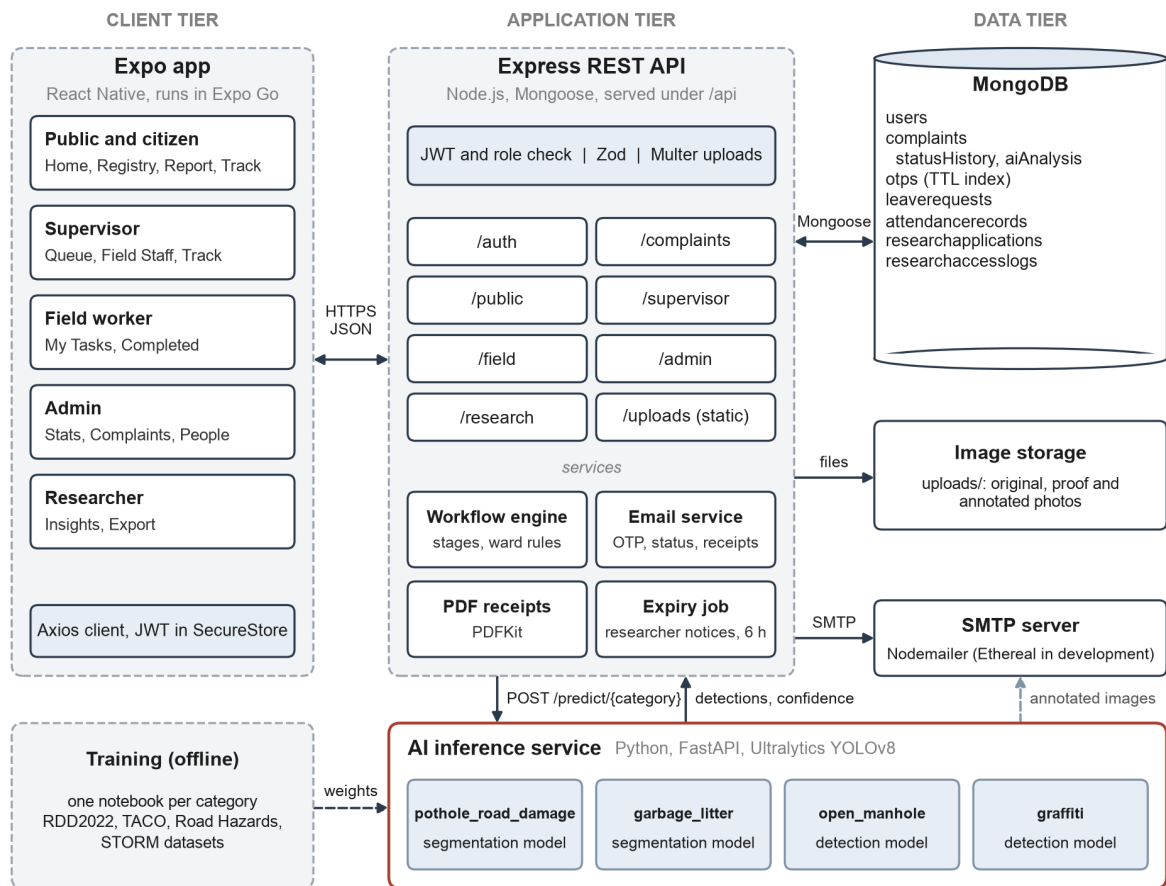


Fig. 4.5 System architecture of UrbanFix

Fig. 4.6 follows one complaint to the supervisor's queue; the AI call runs after the tracking ID is returned.

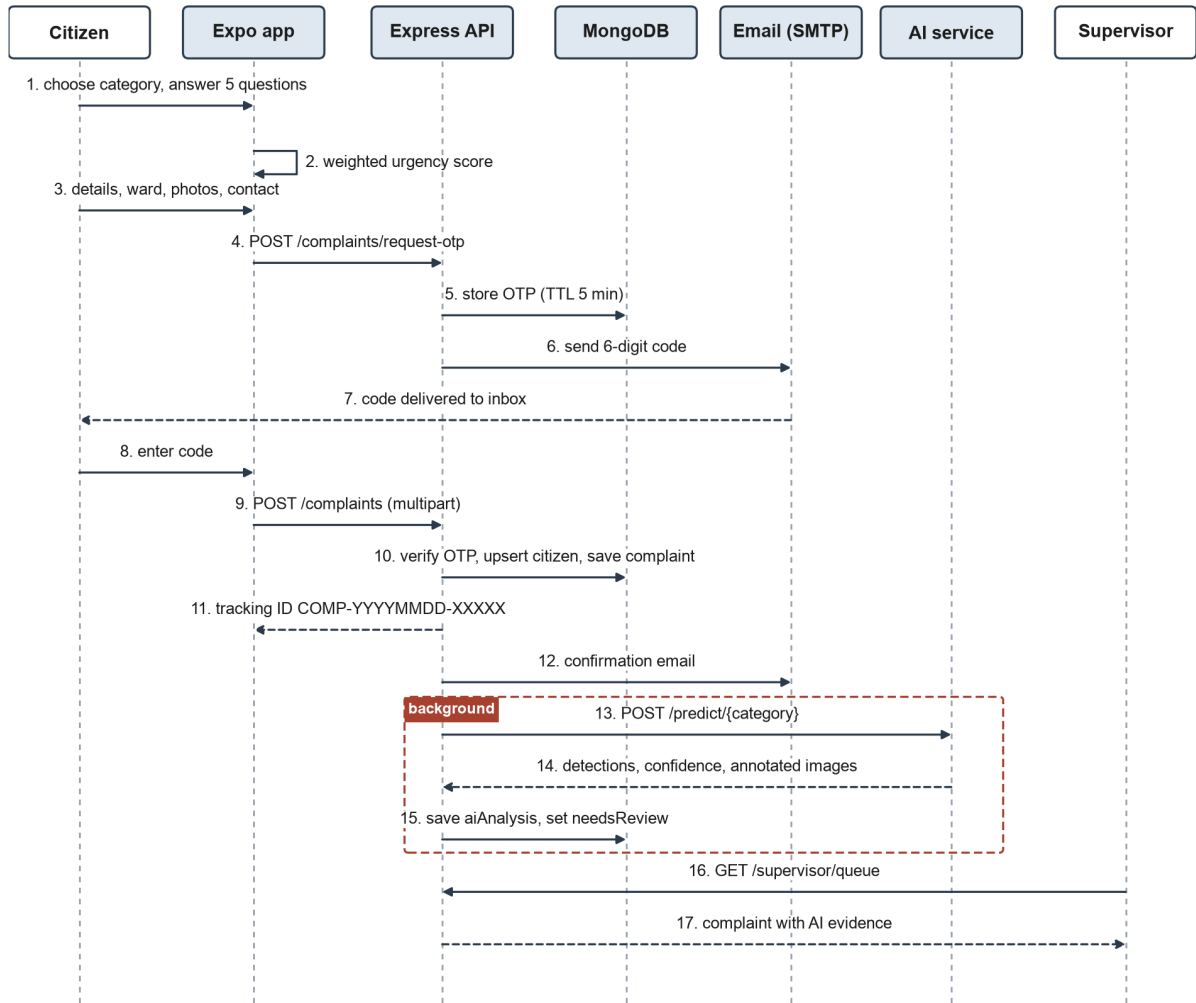


Fig. 4.6 Sequence of complaint submission, OTP verification and AI validation

4.4 Subsystem Services

In the data model (Fig. 4.7) `User` holds every role, and each `Complaint` embeds its `statusHistory` and `aiAnalysis`.

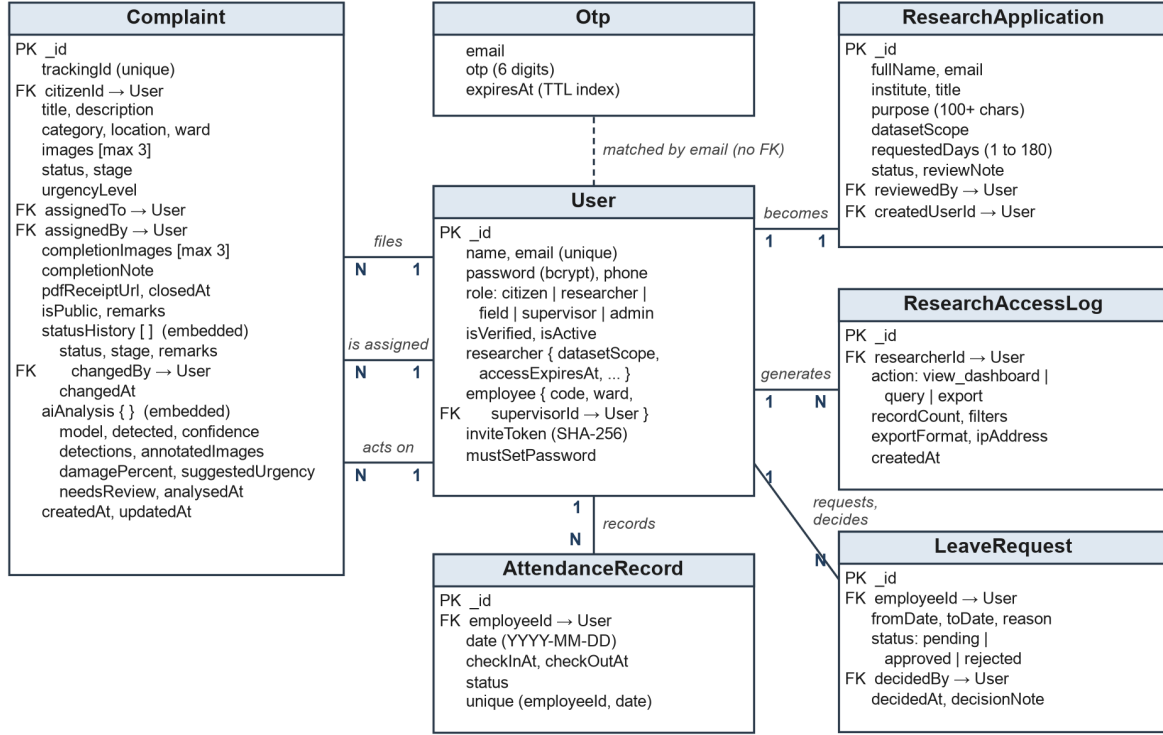


Fig. 4.7 Entity-relationship diagram of the UrbanFix database

The backend also sends emails, builds PDF receipts and warns researchers before their access ends.

4.5 User Interface Designs

Each role has three to five bottom tabs, and questions appear one at a time on swipe cards. Section 5.3 shows the screens.

4.6 Novelty of the Approach

- Categories follow dataset availability, so every report can be checked.
- The model checks the defect itself, which no reviewed system does.
- Urgency is traceable, and a person makes the final call.
- One system covers every step from intake to the closing receipt, plus research access.
- Citizens need no account.

4.7 Summary

The server enforces each step, and every change goes into a public audit trail.

CHAPTER 5

TECHNICAL IMPLEMENTATION AND ANALYSIS

5.1 Outline

This chapter shows the main code, the screens, the tests and the model results.

5.2 Technical Coding and Code Solutions

The urgency score (`mobile/src/utils/urgency.js`) is a weighted share of "Yes" answers, which keeps categories with different question mixes comparable:

```
const HIGH_THRESHOLD = 0.6;
const MEDIUM_THRESHOLD = 0.3;
export const calculateUrgency = (answers, questions = []) => {
  const totalWeight = questions.reduce((sum, q) => sum + (q.weight || 1), 0);
  const scoredWeight = questions.reduce(
    (sum, q) => sum + (answers[q.id] === 'Yes' ? q.weight || 1 : 0), 0);
  return urgencyFromRatio(totalWeight > 0 ? scoredWeight / totalWeight : 0);
};
```

A single object in `backend/utils/complaintStage.js` defines every legal move, and the workflow service rejects anything else. In this listing SUP stands for the supervisor and admin roles:

```
export const TRANSITIONS = {
  accept: { from: ['submitted'], to: 'accepted', roles: SUP, remarks: true },
  reject: { from: ['submitted'], to: 'triage_rejected', roles: SUP, remarks: true },
  assign: { from: ['accepted', 'assigned'], to: 'assigned', roles: SUP },
  start: { from: ['assigned'], to: 'work_in_progress', roles: ['field'], assignedOnly: true },
  proof: { from: ['work_in_progress'], to: 'proof_submitted', roles: ['field'], assignedOnly: true },
  close: { from: ['proof_submitted'], to: 'closed', roles: SUP, remarks: true },
  rework: { from: ['proof_submitted'], to: 'assigned', roles: SUP, remarks: true },
};
```

The API accepts only the four categories that have datasets (`backend/validators/complaintValidator.js`):

```
import { CATEGORIES } from '../constants/categories.js';
category: z.enum(CATEGORIES, { errorMap: () => ({ message: 'Please choose a valid category' }) }),
```

Every model was trained with the same Ultralytics call on the Apple MPS backend, and the inference service runs the saved weights on each photo:

```
model = YOLO('yolov8n-seg.pt' if task == 'segment' else 'yolov8n.pt')
model.train(data=data_yaml, epochs=60, imgsz=640, batch=16, patience=15, device='mps', seed=0)

result = YOLO('weights/pothole_road_damage_model.pt').predict(photo, imgsz=640, conf=0.25)[0]
boxes = result.boxes.xyxy.tolist(); conf = result.boxes.conf.tolist()
```

5.3 Working Layout of Forms

The screens below come from the running app on a demonstration database.

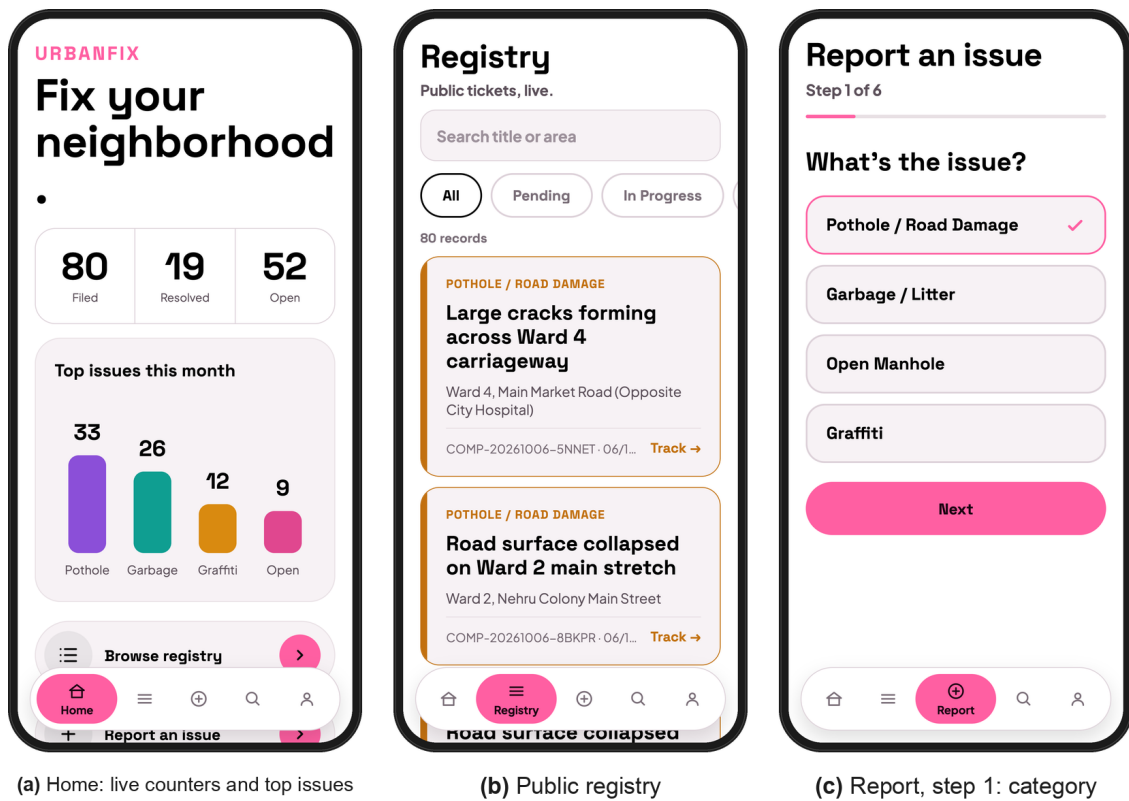


Fig. 5.1 Citizen screens: home, public registry and category selection

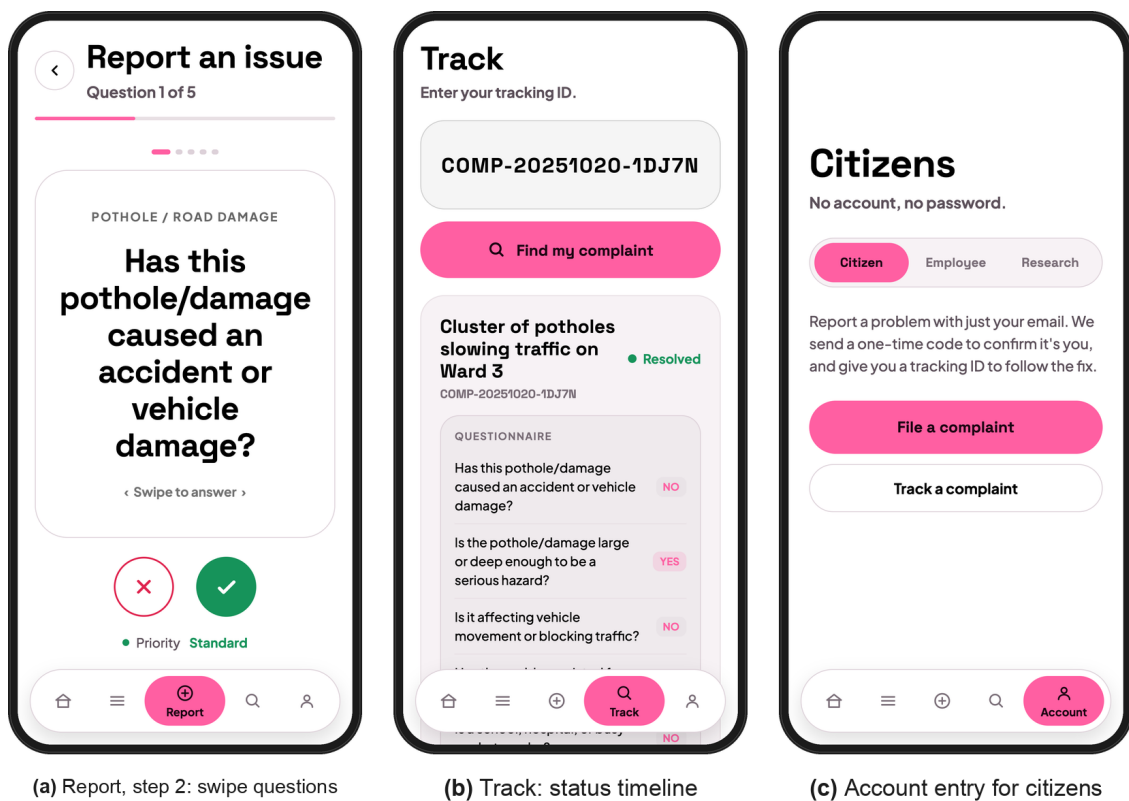


Fig. 5.2 Citizen screens: swipe questionnaire, complaint tracker and account entry

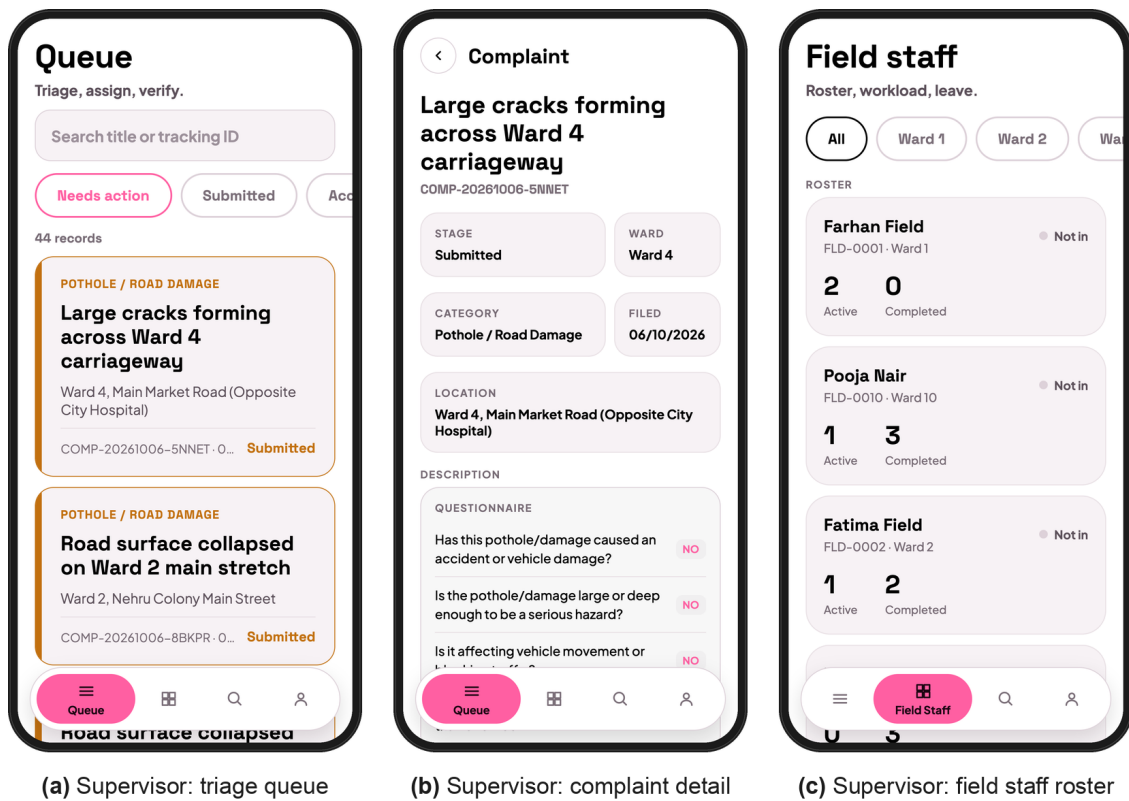


Fig. 5.3 Supervisor screens: triage queue, complaint detail and field staff roster

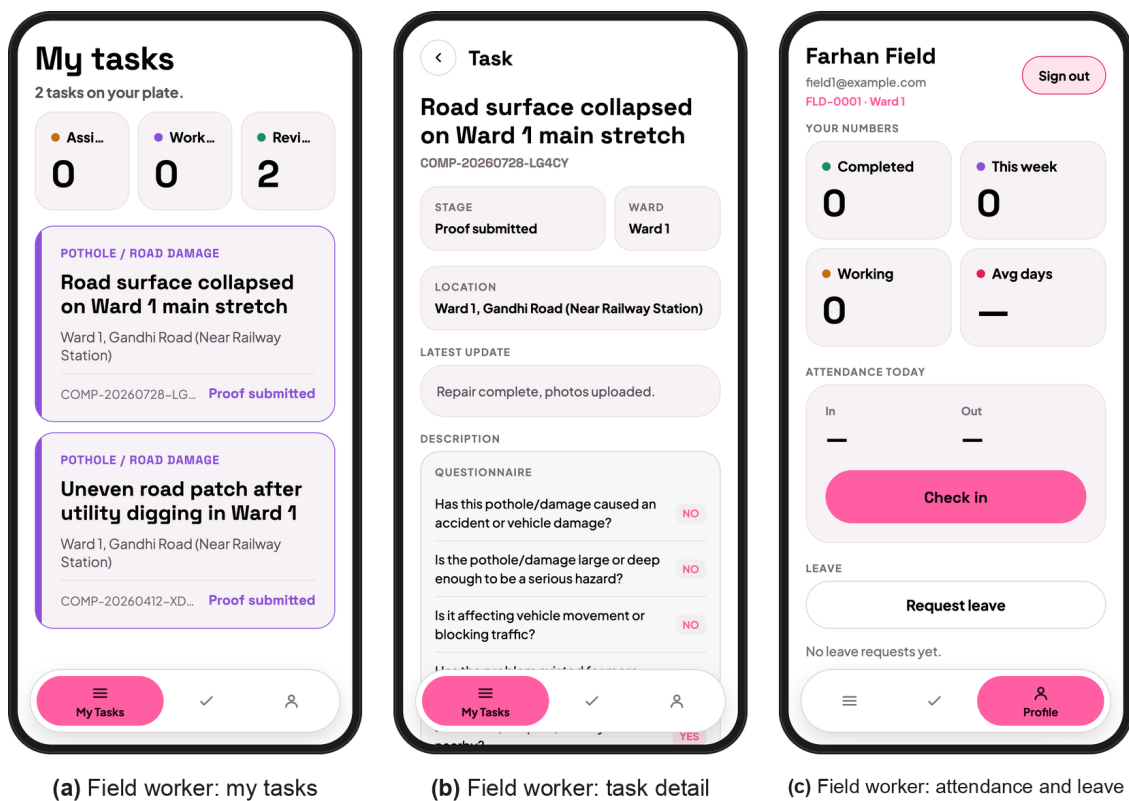


Fig. 5.4 Field worker screens: task list, task detail, attendance and leave

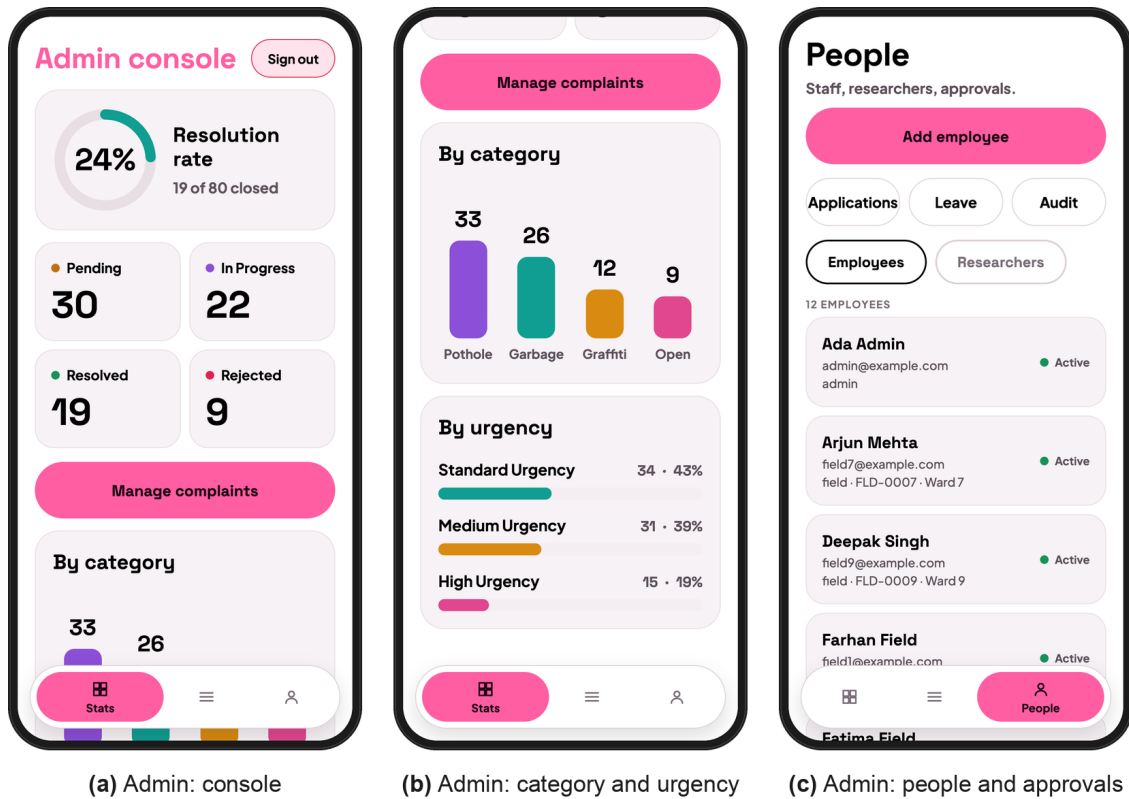


Fig. 5.5 Admin screens: console, category and urgency analytics, people management

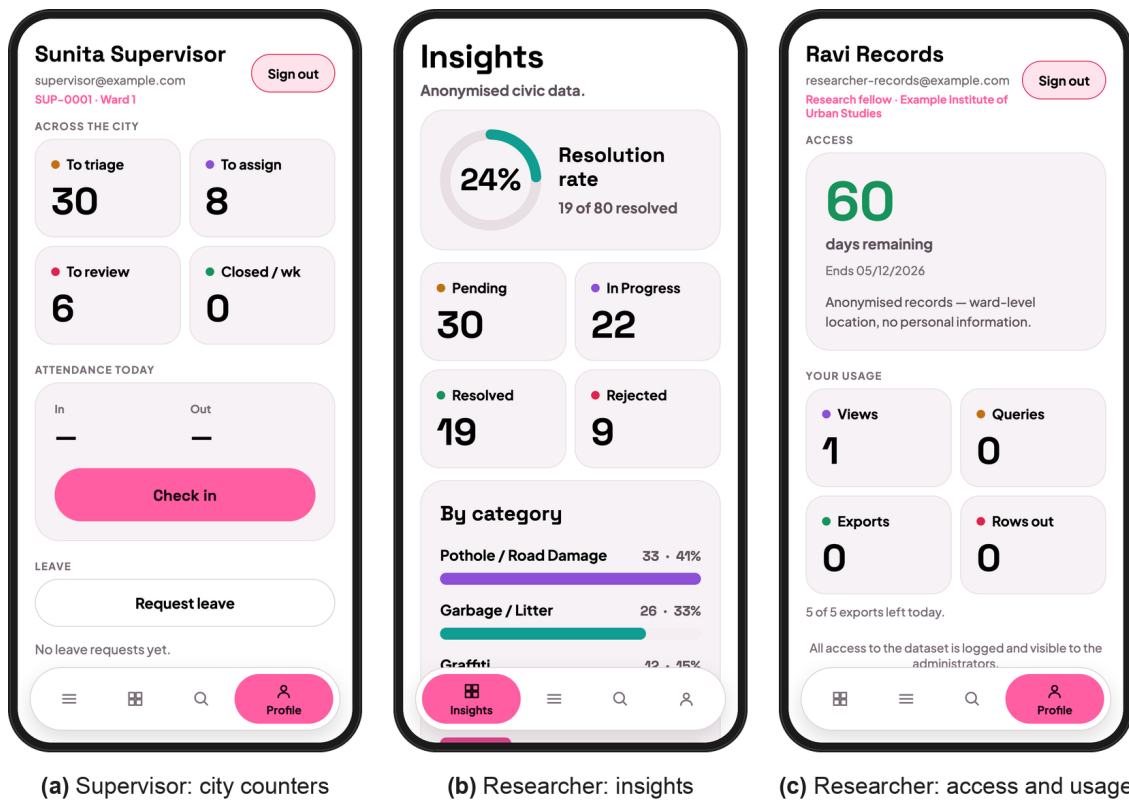


Fig. 5.6 Supervisor city counters and researcher screens: insights, access and usage

5.4 Prototype Submission

The repository has `mobile/`, `backend/` and `ai_model/`. Run `npm run seed:roles` and `npm run dev` in `backend/`, then `npx expo start --go -c` in `mobile/` and scan the QR code with Expo Go.

5.5 Test and Validation

A test script called the running API on a separate database (Table 5.1).

Table 5.1 API validation tests and results

ID	Test	Expected	Observed	Result
T1	Health check	200	200	Pass
T2	OTP request without email	400	400	Pass
T3	Category outside the four datasets	400	400 Please choose a valid category	Pass
T4	Wrong OTP	400	400 Invalid or expired OTP	Pass
T5	Expired OTP	400	400 OTP has expired. Please request a new one.	Pass
T6	Valid submission returns tracking ID	201, COMP-YYYYMMDD-XXXXX	201 COMP-20261006-NKKGR	Pass
T7	Public tracker hides citizen identity	actor shown as "Citizen"	200 Citizen; citizenId absent	Pass
T8	Registry entries carry no citizenId	no citizenId field	200, 0 with citizenId	Pass
T9	Field worker cannot accept	403	403	Pass
T10	Remarks under 10 characters	400	400 Remarks must be at least 10 characters long	Pass
T11	Supervisor accepts with remarks	200, stage accepted	200 accepted	Pass
T12	Reject after accept (illegal transition)	400	400 Cannot reject a complaint that is "Accepted".	Pass
T13	Assign Ward 1 complaint to Ward 2 worker	400	400 This complaint is in Ward 1; Fatima Field covers Ward 2.	Pass
T14	Receipt before closure	400	400 Receipts can only be compiled for resolved complaints.	Pass
T15	Aggregate-only researcher asks for records	403	403 Your access is limited to aggregate statistics.	Pass
T16	Login with wrong password	401	401	Pass

All 16 tests passed.

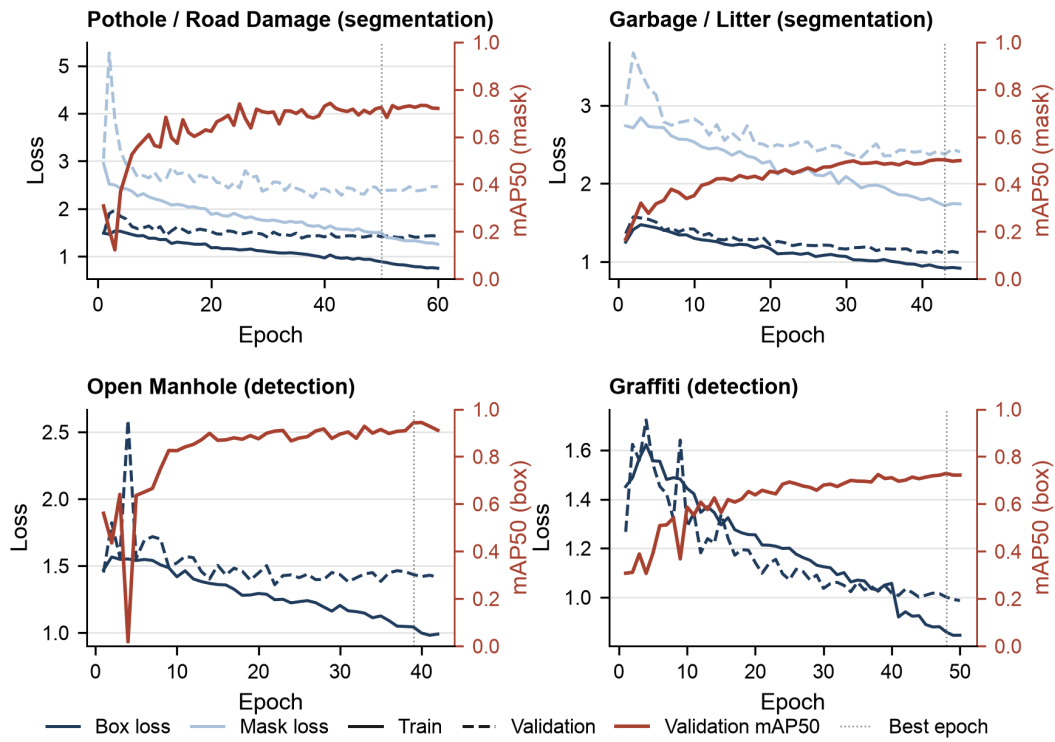
5.6 Performance Analysis

All four models start from COCO-pretrained YOLOv8 nano weights (Table 5.2).

Table 5.2 Model training configuration

Setting	Value
Base models	yolov8n-seg (pothole, litter), yolov8n (open manhole, graffiti)
Image size, batch	640 x 640 pixels, batch 16
Epochs, early stopping	60 maximum, patience 15
Optimiser	Ultralytics auto (AdamW or SGD chosen from iteration count)
Augmentation	Ultralytics defaults: mosaic, HSV jitter, flips, scale, translate
Hardware	Apple M5, PyTorch 2.14 with MPS backend, seed 0

Training took from 29 to 52 minutes per model. Only 1428 of the 1,500 TACO images could still be downloaded.

**Fig. 5.7** Training and validation loss with validation mAP50 for the four category models

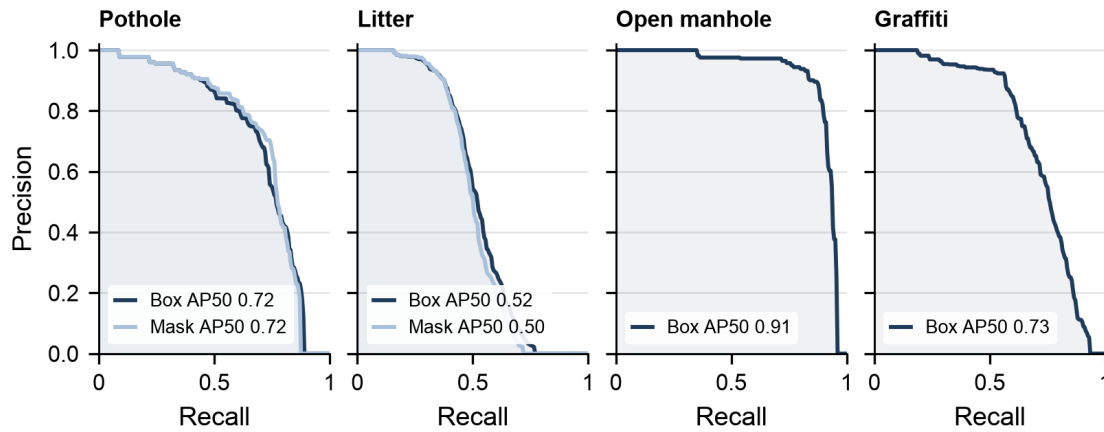
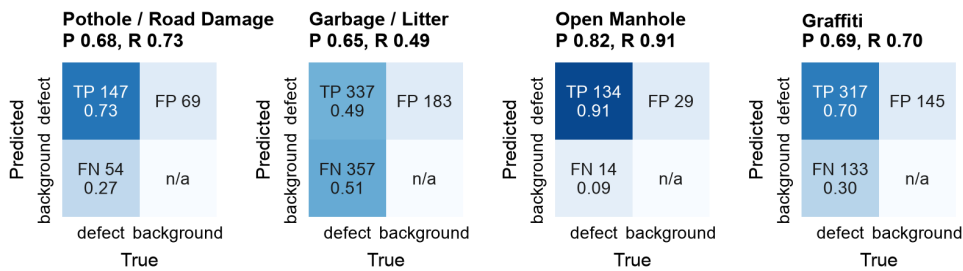


Fig. 5.8 Precision-recall curves on the validation sets



Instance counts at confidence 0.25, IoU 0.45. The defect column is normalised by true defects. Detectors have no background-background cell.

Fig. 5.9 Normalised confusion matrices on the validation sets

Table 5.3 Validation results of the best checkpoint of each model

Model	Task	Precision	Recall	mAP50	mAP50-95
Pothole / Road Damage (box)	segment	0.733	0.682	0.719	0.447
Pothole / Road Damage (mask)	segment	0.725	0.711	0.724	0.426
Garbage / Litter (box)	segment	0.748	0.446	0.521	0.375
Garbage / Litter (mask)	segment	0.757	0.434	0.502	0.317
Open Manhole (box)	detect	0.890	0.874	0.908	0.469
Graffiti (box)	detect	0.836	0.604	0.729	0.526

Open manholes score highest (mAP50 0.91) thanks to their regular shape, though augmented copies in the dataset may inflate this. The pothole model reaches 0.72, matching the earlier notebook run. Graffiti is precise (0.84) but misses faint tags (recall 0.60), and litter is hardest (0.50) because TACO merges 60 object types into one class. When the model misses a defect, the complaint goes to a supervisor for review.

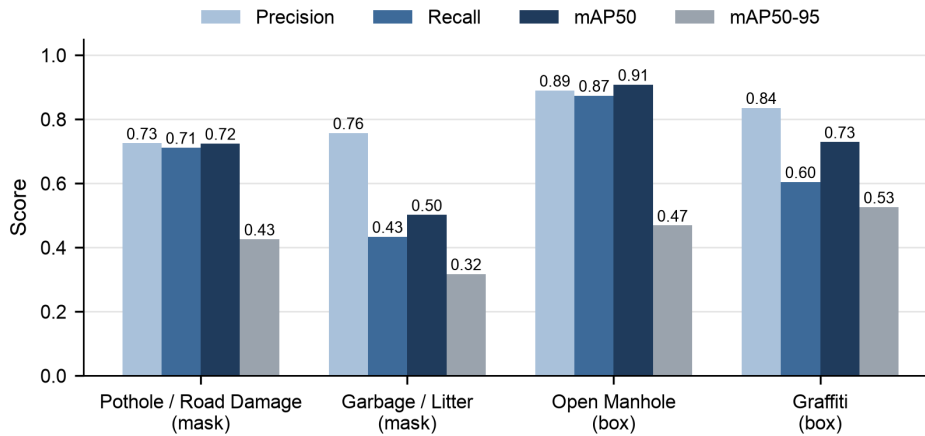


Fig. 5.10 Precision, recall and mAP50 of the four models side by side

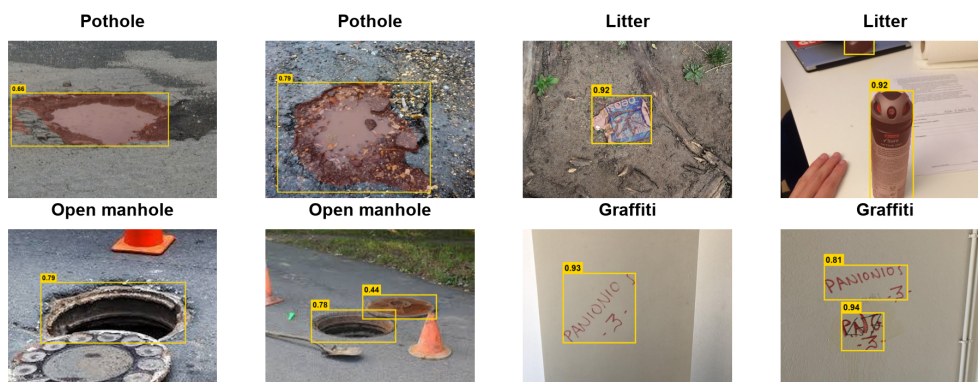
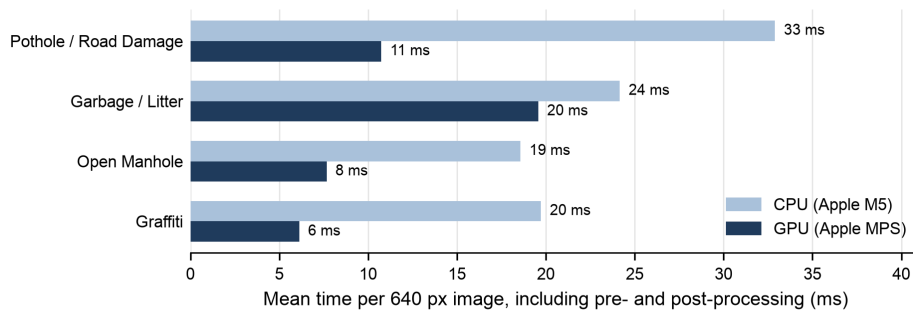


Fig. 5.11 Sample validation images with model output for each category



Mean over 40 validation images after 3 warm-up runs, batch size 1.

Fig. 5.12 Inference time per image on CPU and on the Apple MPS GPU backend

A photo takes 19 to 33 ms on the laptop CPU (Fig. 5.12), so the AI service needs no GPU server.

5.7 Summary

All tests pass, and the models are accurate and fast enough for ward-level use on a laptop CPU.

CHAPTER 6

PROJECT OUTCOME AND APPLICABILITY

6.1 Outline

What UrbanFix delivers and where it could be used.

6.2 Key Implementation Outlines of the System

- An Expo app with five tab sets, one per role, that runs on Android and iOS through Expo Go.
- An Express API with seven route groups, workflow rules enforced on the server, invite-based onboarding and an audit trail on every complaint.
- Four YOLOv8 models, one per category, trained on public datasets and served through a FastAPI inference service.
- A public registry and tracker, PDF receipts, and a research layer that protects privacy, logs access and enforces limits.

6.3 Significant Project Outcomes

We seeded 80 demonstration complaints over twelve months and moved them through the stages (Figs. 6.1 and 6.2). Because the data is generated, the charts illustrate the dashboards and do not measure a real ward.

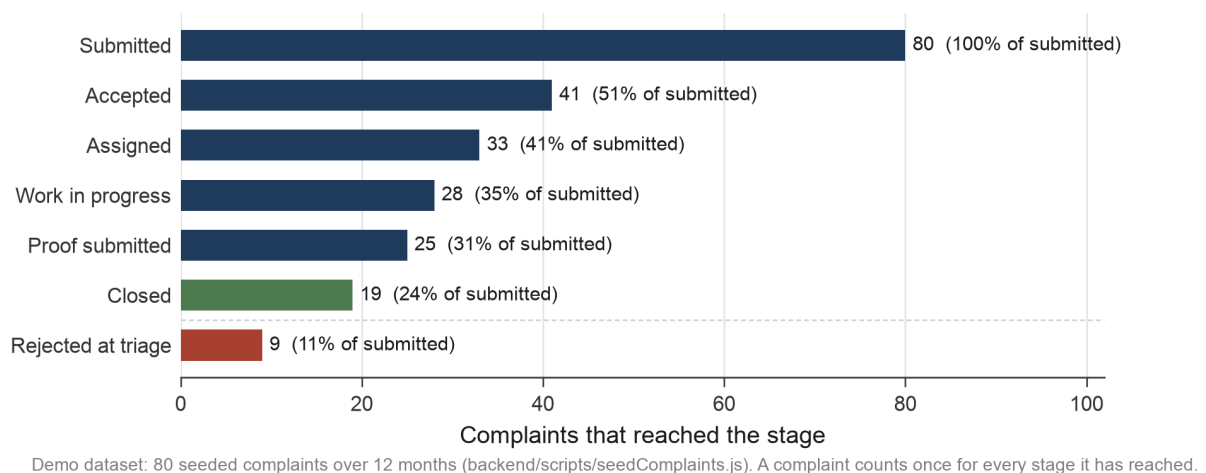
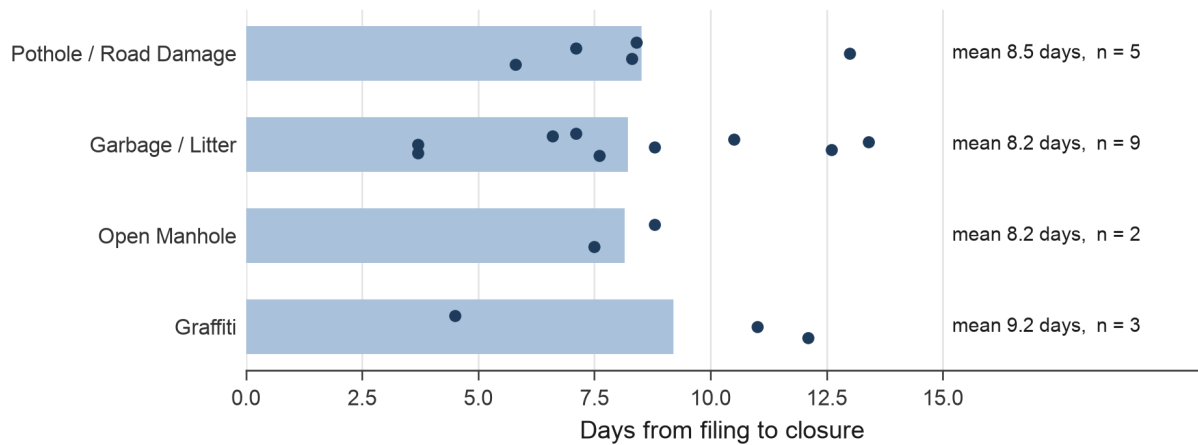


Fig. 6.1 Complaints reaching each workflow stage (demonstration dataset)



Demo dataset (seeded). Bars show the mean; dots show individual closed complaints.

Fig. 6.2 Days from filing to closure by category (demonstration dataset)

6.4 Project Applicability on Real-World Applications

- A municipal corporation could run UrbanFix ward by ward, mapping supervisors to wards and field staff to teams.
- A Smart City SPV could publish the registry and resolution data as an open performance dashboard.
- Researchers and planners could study complaint patterns by category, ward and month using anonymised exports.
- Campuses, townships and housing societies could use the same workflow for their own maintenance.

6.5 Inference

A student team built a full governance loop with open tools and public data.

CHAPTER 7

CONCLUSIONS AND RECOMMENDATION

7.1 Outline

UrbanFix checks, prioritises, routes and publishes civic complaints in one system.

7.2 Limitations and Constraints of the System

- Only four categories have suitable public datasets, so common complaints such as water leakage and faulty streetlights are left out.
- The models learned from public data collected in other cities and countries, and nobody has measured their accuracy on Indian street photos in different light and weather.
- We chose the question weights ourselves; they have not been fitted to real outcomes.
- Location is recorded as a ward and free text rather than GPS, and there is no duplicate detection, so a pothole reported twice appears twice.
- The workflow results in Chapter 6 come from demonstration data, there has been no pilot with a municipal body, and citizens need a smartphone, internet access and email.

7.3 Future Enhancements

- Collect and label local photos to fine-tune each model, and add categories as new public datasets appear.
- Record GPS with each report, add map views and duplicate detection, and learn the question weights from closed complaints.
- Support Hindi and regional languages, and add an SMS or IVR channel for citizens without smartphones.
- Pilot the system in one ward of a municipal corporation and compare its accuracy with manual review, as CivicX [6] did.

7.4 Inference

No reviewed system combines defect-level photo checks, traceable urgency and a public record. The next step is a field pilot.

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APPENDIX A: API ENDPOINTS

Table A.1 REST API route groups

Prefix	Who	Endpoints
/api/auth	everyone	POST /register, /verify-otp, /resend-otp, /login, /set-password; GET /me
/api/complaints	citizens	POST /request-otp, / (multipart); GET /track/:id, /download-receipt/:id, /my-complaints
/api/public	anyone	GET /complaints (category, status, ward, location, page), /stats
/api/supervisor	supervisor, admin	GET /queue, /complaints/:id, /field-staff, /stats, /leave; PATCH /complaints/:id/accept, reject, assign, close, rework; PATCH /leave/:id
/api/field	field worker	GET /tasks, /tasks/:id, /stats, /leave, /attendance; PATCH /tasks/:id/start; POST /tasks/:id/proof, /leave, /attendance/check-in, /attendance/check-out
/api/admin	admin	GET /stats, /activity-heatmap, /complaints, /users, /research-applications, /audit/research, /leave; POST /users; PATCH /complaints/:id/status, /users/:id, deactivate, reactivate, /research-applications/:id, /leave/:id
/api/research	researcher, admin	POST /apply (public); GET /me, /dashboard, /query, /export?format=csv json
AI service	backend only	POST /predict/{category}; GET /models, /health